

The Backend phase transforms your architecture and requirements into detailed API specifications, business logic, and data management strategies. This section focuses on creating comprehensive backend specifications that power your chatbot application with robust, scalable, and secure server-side functionality.

What You'll Learn

In this backend phase, you'll:

- **Design comprehensive APIs** with OpenAPI specifications and RESTful patterns
- **Implement business logic** with service layer architecture and domain modeling
- **Create data management** strategies with efficient storage and retrieval patterns
- **Build integrations** with external services, AI models, and third-party APIs

Backend Exercises

1. API Design and Specification

Create comprehensive API specifications using OpenAPI standards, including endpoint design, request/response schemas, authentication, and error handling.

2. Business Logic and Service Architecture

Design service layer architecture with domain modeling, business rules implementation, and clean separation of concerns.

3. Data Management and Storage

Implement data management strategies including database design, caching, data access patterns, and performance optimization.

4. External Integrations and Services

Create integration specifications for Amazon Bedrock, third-party APIs, webhooks, and event-driven architectures.

Key Benefits

By completing the Backend phase, you'll have:

- **Comprehensive API Specifications:** Well-documented APIs with clear contracts
- **Scalable Business Logic:** Clean architecture with maintainable business rules
- **Efficient Data Management:** Optimized data storage and retrieval strategies
- **Robust Integrations:** Reliable connections to external services and AI models

Advanced Features Integration

Throughout this phase, you'll leverage:

- **AWS Documentation MCP Server:** For current AWS service best practices
- **Web Search MCP Server:** For API design patterns and backend architectures
- **Agent Hooks:** For API validation and documentation generation
- **Agent Steering:** For backend coding standards and architectural patterns

Prerequisites

Before starting the Backend exercises:

- Foundation and Infrastructure phases completed
- Understanding of RESTful API design principles
- Familiarity with serverless architecture patterns
- Knowledge of database design and data modeling

Let's begin with comprehensive API design that creates clear contracts between your frontend and backend systems while following industry best practices and standards.